Introduction

Brief Recap of the Data Set:

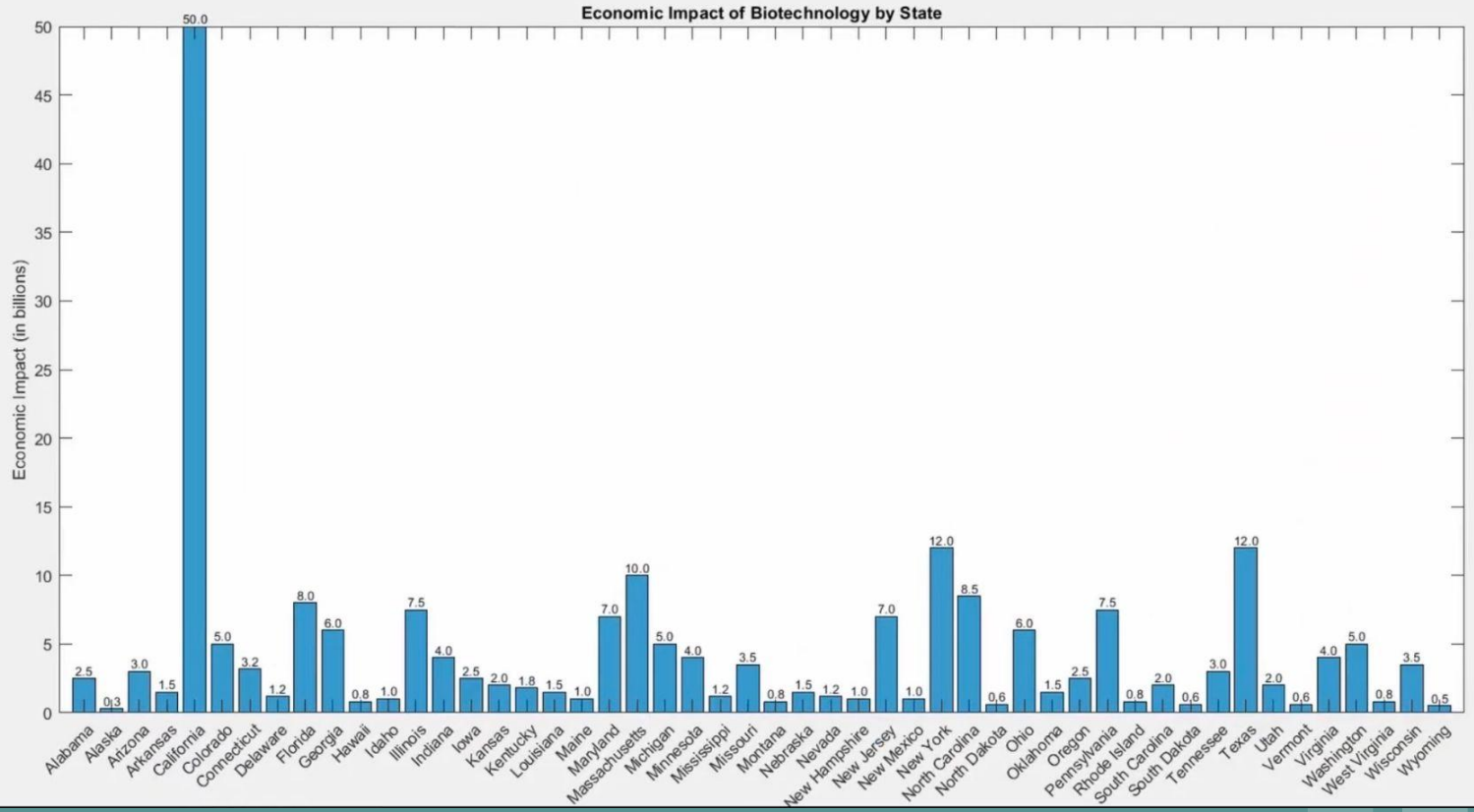
- **Data Aggregation and Cleaning**: Gathered and standardized publicly available biotechnology data for all 50+ U.S. states.
- **Algorithm Development**: MATLAB script to analyze trends and create visualizations for employment, economic impact, and regional biotech strengths.
- **Attachments**: Please refer to the attached files related to this project for MATLAB.

Goals and Tasks

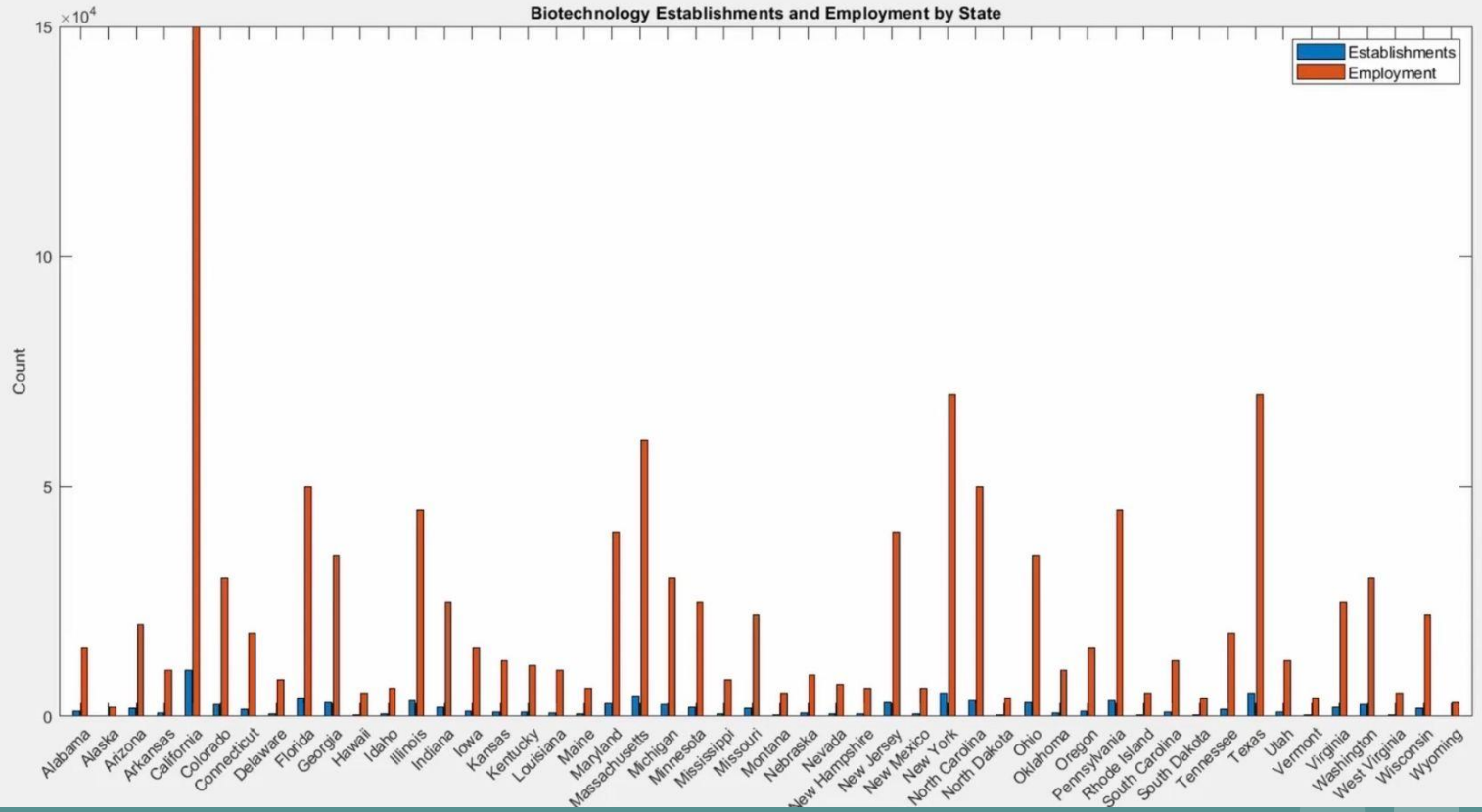
Goal: Which U.S. states are leading in biotechnology innovation and economic impact, and which regions have untapped potential for growth in the biotech sector?

Tasks: The correlations presented in MATLAB are our hypothesis and through these data visualizations we will discover the relationship between employment, economic impact, and regional biotech strengths.

Economic Impact of Biotechnology by State



Biotechnology Establishments and Employment by State



Summary and Conclusions

Biotechnology Industry Summary by State

- California leads the biotechnology industry with the highest economic impact (~$50B), employment, and number of establishments.
- Other major biotech hubs include Texas, New York, Massachusetts, Pennsylvania, and North Carolina.
- States with stronger research universities, healthcare systems, and investment ecosystems show greater biotech growth.
- Economic impact, employment, and number of establishments are strongly correlated across states.
- Smaller states generally show limited biotechnology activity due to lower infrastructure and investment.

Conclusion

The biotechnology industry is heavily concentrated in a few innovation-driven states, demonstrating the importance of research funding, skilled workforce development, and strong innovation ecosystems in driving economic growth. In future iterations, I would simplify the graphs further and improve label readability.